

Development and Study of New Deep Learning Architectures for Skin Disease Classification

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A thesis submitted in group fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering



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Declaration

This is to certify that the thesis work entitled **Development and Study of new Deep Learning Architectures for Skin Disease Classification** has been carried out by **Abu Bakar Siddique**, Student ID: 12221003, **Onik Howlader**, Student ID: 12221004 **Mizanur Rahman**, Student ID: 12221044, and **Md. Hasibul Islam Khan**, Student ID: 12221053 in the Department of Computer Science and Engineering, University of Global Village (UGV), Barishal, Bangladesh. The thesis work or any portion of it has never been submitted to a university or other institution for the granting of a degree or diploma.

Signature of Supervisor

Signature of Candidates

Approval

This is to certify that the thesis submitted by Abu Bakar Siddque, Onik Howlader, Mizanur Rahman and Hasib Khan entitled *Development and Study of new Deep Learning Architectures for Skin Disease Classification* has been approved by the board of examiners in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering in the Department of Computer Science and Engineering, University of Global Village, Barishal, Bangladesh.

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Abstract

Skin diseases are common and delays in diagnosis can lead to ongoing symptoms, discomfort and more complications. Detecting these conditions early is important for better patient outcomes especially where there are few dermatologists. Computer aided analysis of skin images helps with early screening and highlights cases that need more clinical attention. These tools are meant to support not replace clinical expertise. In this study, we developed and tested four deep-learning models: ABS Hybrid, UgvNet-DS, TRY NET-8 and UGVNet. We also compared them to three baseline models: Conventional CNN, VGG16, and ResNet50. We used the SkinDisNet dataset which includes 1,710 clinical images from 416 patients and covers Atopic Dermatitis (AD), Contact Dermatitis (CD), Eczema (EC), Scabies (SC), Seborrheic Dermatitis (SD) and Tinea Corporis (TC). The dataset was split into 1,197 training images, 256 validation images and 257 test images (70:15:15 split). We applied offline data augmentation only to the training set increasing it to 9,576 images while the validation and test sets stayed the same to avoid data leakage. To handle class imbalance and the visual similarity between diseases, we trained the models using class balanced focal loss and evaluated them with accuracy, macro-averaged precision, recall, F1-score and class-wise metrics of the new models, UGVNet performed best. It combines EfficientNetV2-S and ConvNeXt-Tiny backbones with adaptive gated feature fusion and 8-head global self-attention. UGVNet reached 91.83% test accuracy and a macro-F1 score of 0.9250, correctly identifying all minority Atopic Dermatitis cases in the test set. ABS Hybrid, which uses EfficientNetV2-M and ConvNeXt-Base achieved 90.66% accuracy and a macro-F1 score of 0.8877. TRY NET-8, a lightweight model built from scratch reached 85.21% accuracy and a macro-F1 score of 0.8170 with only 3.49 million parameters (about 13.95 MB) matching the accuracy of the much larger ResNet50 baseline. The earlier UgvNet-DS version achieved 90.27% accuracy and a macro-F1 score of 0.8919 further supporting model development. Overall, the new models outperformed the baseline models showing their potential to improve automated skin disease classification and help with clinical decision making.

Keywords: Skin Disease Classification, Deep Learning, Medical Image Analysis, Computer Aided Diagnosis, SkinDisNet, Feature Fusion, Self Attention, Focal Loss, UGVNet.

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List of Abbreviations

AD	Atopic Dermatitis
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CD	Contact Dermatitis
CNN	Convolutional Neural Network
ConvNeXt	Modernized Pure Convolutional Neural Network
EC	Eczema
ECE	Expected Calibration Error
FLOPs	Floating-Point Operations
FPN	Feature Pyramid Network
GAP	Global Average Pooling
GELU	Gaussian Error Linear Unit
GMP	Global Max Pooling
HAM10000	Human Against Machine with 10,000 Training Images Dataset
ISIC	International Skin Imaging Collaboration
MBConv	Mobile Inverted Bottleneck Convolution
MLP	Multi-Layer Perceptron
ReLU	Rectified Linear Unit
SC	Scabies
SD	Seborrheic Dermatitis
SE	Squeeze-and-Excitation
Swin	Shifted Window (Hierarchical Vision Transformer)
TC	Tinea Corporis
UGV	University of Global Village
UGVNet	Unified Global Vision Network
VGG	Visual Geometry Group
ViT	Vision Transformer

Chapter I

Introduction

1.1 Background

Skin diseases represent a major global health burden impacting both physical well being and psychological quality of life [1]. Developing automated computer vision tools for dermatological image analysis offers significant potential for clinical decision support. However, skin condition classification is inherently challenging due to high inter-class visual mimicry across distinct pathologies and substantial intra-class morphological variation in color, texture, lesion margins, and acquisition conditions.

Deep learning has transformed medical image analysis by learning hierarchical visual representations directly from pixel data [2]. Seminal investigations demonstrated the viability of convolutional neural networks (CNNs) across clinical and dermoscopic datasets [3, 4], underscoring the critical role of benchmark composition and rigorous evaluation protocols.

Modern vision architectures present distinct representational and computational trade-offs: VGG relies on deep sequential convolutions [5], ResNet introduces residual skip connections [6], and EfficientNet utilizes compound scaling [7, 8]. Recent innovations such as ConvNeXt [9] and hierarchical vision transformers like Swin Transformer V2 [10] further broaden feature-extraction paradigms. Hybrid architectures seek to unite local convolutional inductive bias with global contextual attention. However, increased architectural complexity does not inherently ensure superior generalization and often incurs substantial computational overhead. Systematic benchmarking under a unified protocol is therefore essential.

This study investigates seven deep learning configurations on the six-class SkinDisNet clinical benchmark [11, 12]:

- **Baselines:** Conventional CNN (trained from scratch), VGG16, and ResNet50.
- **Proposed Models:** ABS Hybrid (EfficientNetV2-M + ConvNeXt-Base), UgvNet-DS (DenseNet121 + Swin Transformer V2), TRY NET-8 (Attentive Residual MBConv

CNN), and UGVNet (EfficientNetV2-S + ConvNeXt-Tiny with adaptive gated fusion and global self-attention).

1.2 Motivation

The motivation for this research is twofold:

1. **Clinical and Methodological Fairness:** Severe but rare skin diseases can be missed due to false negatives. Macro averaged metrics, class-wise recall, and confusion patterns help assess minority class performance without majority class bias.
2. **Architectural and Practical Efficiency:** High capacity multi-backbone models require substantial memory and computation, limiting deployment in resource-constrained healthcare settings. Comparing them with lightweight networks helps determine whether their added complexity provides meaningful diagnostic benefits.

1.3 Problem Statement

Automated skin disease classification faces two key challenges: class imbalance and visual similarity between diseases. Accuracy alone may hide poor minority class performance, while the best balance between accuracy, sensitivity and computational efficiency across lightweight and hybrid models remains unclear.

This study asks: Under a standardized dataset and evaluation protocol, which deep learning model provides the best balance of diagnostic performance, class wise sensitivity, and computational efficiency for six-class skin disease classification?

1.4 Objectives

The primary objective of this thesis is to conduct a rigorous comparative evaluation of three baseline architectures and four proposed custom/hybrid models on the SkinDisNet benchmark.

The specific research objectives are:

1. Partition the dataset into six disease specific subsets and audit each for data integrity, class balance, and consistent stratification across the training, validation, and test sets.

2. Implement and evaluate three baseline architectures: Conventional CNN, VGG16, and ResNet50.
3. Design, implement, and benchmark four proposed architectures: ABS Hybrid, UgvNet-DS, the compact from-scratch TRY NET-8, and UGVNet.
4. Assess all configurations on identical fixed train, validation, and test partitions using multi-faceted evaluation metrics including precision, recall, macro-F1, balanced accuracy, and confusion dynamics.
5. Analyze the impact of class imbalance on minority categories (Atopic Dermatitis and Seborrheic Dermatitis) and evaluate class-balanced mitigation strategies.
6. Characterize computational complexity across models in terms of parameter counts, input dimensions, and memory footprint.

1.5 Organization of the Thesis

This thesis is structured into five cohesive chapters:

- **Chapter II (Background Study and Related Works):** Reviews theoretical foundations of convolutional networks, attention mechanisms, residual learning, hybrid paradigms, related clinical benchmarks, and identifies the research gap.
- **Chapter III (Methodology):** Details dataset partitioning, training-only augmentation, loss formulations, architectural designs of all seven models, and the evaluation framework.
- **Chapter IV (Results and Discussion):** Presents empirical training convergence, test-set classification reports, class-wise sensitivity, confusion analyses, complexity trade-offs, and threats to validity.
- **Chapter V (Conclusion and Future Work):** Summarizes key findings, discusses practical trade-offs, and outlines future research directions.

Chapter II

Background Study and Related Works

2.1 Background Study

This section reviews the concepts needed for the seven-model comparison: dermatological image classification, transfer learning, residual and efficient convolution, attention, multi scale representation, and multi-backbone fusion. Their roles in the evaluated architectures are specified in Chapter III.

2.1.1 Skin Diseases and Dermatological Image Classification

Skin diseases impose a major global non-fatal burden [1]. Automated classification remains difficult because different dermatological conditions may share similar visual characteristics, while the appearance of the same disease can vary according to skin tone, body site, disease stage, lighting conditions, and image quality.

Clinical, dermoscopic, and histopathological images represent different imaging modalities, and performance obtained from one modality does not necessarily transfer directly to another [3, 4]. In addition, dermatological datasets may contain class imbalance, repeated lesions, site bias, and incomplete metadata [13]; classification performance may also vary across demographic groups [14].

The present study focuses on six dermatological disease classes contained in the SkinDisNet dataset. A brief background of each disease category is provided below.

Atopic Dermatitis (AD)

Atopic Dermatitis (AD) is a chronic inflammatory skin condition commonly associated with dry, itchy, irritated, and inflamed skin. Depending on the disease stage and severity, the affected region may appear rough, scaly, reddish, or discolored. Its visual appearance may vary considerably across patients, which can make image-based differentiation from other dermatitis-related conditions challenging.

Contact Dermatitis (CD)

Contact Dermatitis (CD) is an inflammatory skin reaction that develops after exposure to an irritating substance or an allergen. Common visible characteristics include redness, swelling, dryness, scaling, and localized irritation. Because several of these features are also observed in other inflammatory skin diseases, Contact Dermatitis may show considerable visual similarity with Atopic Dermatitis and Eczema.

Eczema (EC)

Eczema (EC) refers to a group of inflammatory skin conditions commonly characterized by itching, dryness, redness, scaling, and changes in skin texture. Its appearance can vary significantly depending on severity, body location, and individual skin characteristics. This intra-class variation, together with similarities to other dermatitis-related conditions, makes automated recognition of Eczema challenging.

Scabies (SC)

Scabies (SC) is a contagious skin condition caused by infestation with the human itch mite. It commonly produces intense itching along with small papules, rashes, or other visible skin lesions. The appearance of Scabies may vary depending on the affected body region and severity, and some cases may resemble inflammatory or eczematous skin conditions in clinical photographs.

Seborrheic Dermatitis (SD)

Seborrheic Dermatitis (SD) is a chronic inflammatory skin disorder that commonly affects areas rich in sebaceous glands, including the scalp, face, and upper body. It is often associated with redness, scaling, flaking, and irritated skin. Its visual characteristics may overlap with Eczema and other forms of dermatitis, which can increase the difficulty of automated image classification.

Tinea Corporis (TC)

Tinea Corporis (TC), commonly known as ringworm of the body, is a superficial fungal infection of the skin. It often appears as a circular or ring-shaped lesion with a relatively prominent

border and may also include redness, scaling, or itching. However, lesion shape and appearance may vary according to disease progression, skin tone, and image conditions, making some cases visually similar to other dermatological disorders.

2.1.2 Deep Learning in Medical Image Classification

Deep learning learns hierarchical representations directly from data: shallow layers capture edges and textures, while deeper layers encode task-specific patterns [2].

Medical-image studies confirm its predictive value but show that results depend on labels, dataset design, and evaluation protocol [3, 4, 15].

Preprocessing, augmentation, class imbalance, checkpoint selection, and test isolation can materially change performance [16].

Accordingly, this study reports overall and class-wise metrics, confusion patterns, and computational cost.

2.1.3 Convolutional Neural Networks

Convolutional neural networks (CNNs) process grid-structured data like images by applying learnable kernels across local neighborhoods with shared weights [17, 18]. For input feature maps $\mathbf{x}$, the activation for channel k at spatial location (i, j) is given by:

$$y_k(i, j) = \sigma \left[b_k + \sum_c \sum_u \sum_v W(k, c, u, v) \cdot x(c, i + u, j + v) \right] \quad (2.1)$$

Here, $\mathbf{W}$ represents convolution kernels, b is the bias term, and σ is a non-linear activation function (such as ReLU, GELU, or Swish). Weight sharing cuts parameters compared to fully connected layers, while local connectivity leverages the spatial prior that adjacent pixels form cohesive visual patterns. Pooling or strided convolutions lower spatial resolution, and repeated blocks expand the effective receptive field.

AlexNet established large-scale CNN classification, and VGG standardized deep stacks of 3×3 kernels [5, 18]. This thesis uses a Conventional CNN and VGG16 as baselines.

CNNs capture local patterns efficiently but may miss global structure and become difficult to optimize when very deep.

2.1.4 Transfer Learning

Transfer learning adapts source-task knowledge to a target task [19]. ImageNet-pretrained models provide reusable low-level features, while deeper features require task-specific fine-tuning because natural and clinical images differ.

Its value depends on normalization, augmentation, trainable layers, schedules, and checkpoint rules.

This study compares pretrained backbones with the from-scratch Conventional CNN and TRY NET-8.

2.1.5 Attention Mechanisms

Attention strengthens informative features and suppresses irrelevant responses [20]. This study uses three forms:

- **SE channel attention** recalibrates channels through global pooling and learned gates [21]:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_{i,j,c}, \quad s = \sigma(W_2 \delta(W_1 z)) \quad (2.2)$$

where δ is ReLU, σ is sigmoid, and $\tilde{X} = s \odot X$.

- CBAM applies channel attention followed by spatial attention [22]:

$$M_s(F) = \sigma \left(f^{7 \times 7}([\text{AvgPool}(F); \text{MaxPool}(F)]) \right) \quad (2.3)$$

- Self-attention models content-dependent interactions between image tokens [10, 20, 23]:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} + B \right) V \quad (2.4)$$

Swin V2 uses shifted windows and continuous position bias B .

UgvNet-DS uses CNN–Swin attention, TRY NET-8 uses SE blocks, and UGVNet uses gated fusion with multi-head self-attention.

2.1.6 Residual Learning

Residual learning improves deep-network optimization by learning a correction $\mathcal{F}(x)$ to the input [6]:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (2.5)$$

The shortcut preserves information and gradient flow; a projection aligns mismatched dimensions. This thesis evaluates ResNet50 and uses residual blocks in TRY NET-8.

2.1.7 Depthwise-Separable Convolution and MBConv

Depthwise-separable convolution splits spatial filtering from 1×1 channel mixing [24, 25]. Its cost falls from $K^2 M N F^2$ to:

$$K^2 \cdot M \cdot F^2 + M \cdot N \cdot F^2 \quad (2.6)$$

with an approximate cost ratio of:

$$\frac{1}{N} + \frac{1}{K^2} \quad (2.7)$$

MBConv and Fused-MBConv extend this design in EfficientNet and EfficientNetV2 [7, 8], supporting the EfficientNet-based components of ABS Hybrid and UGVNet.

2.1.8 Multi-Scale Feature Extraction

Multi-scale extraction combines fine detail with broad context [26, 27]. Inception uses parallel receptive fields, Feature Pyramid Networks merge resolutions, and ConvNeXt and Swin build hierarchical representations [9, 10, 23]. The proposed models obtain multi-scale features through backbone fusion or residual-attention hierarchies.

Table 2.1: Related-work synthesis and relation to the proposed study.

Study	Data / Focus	Contribution	Gap / Relation
Esteva et al. [3]	Skin lesions; CNN	Established deep skin-image classification	Different task and dataset
Liu et al. [4]	Clinical photos; differential diagnosis	Broadened clinical-photo classification	Different labels and protocol
Daneshjou et al. [14]	Diverse clinical images; subgroup analysis	Exposed performance disparities	Supports dataset-specific claims
Sultana et al. [12]	SkinDisNet; three pretrained CNNs	Direct six-class benchmark	No compact or hybrid models
Proposed study	SkinDisNet; seven models	Adds compact/hybrid models and class-wise efficiency analysis	Single dataset; external validation remains

2.1.9 Efficient CNN Architectures

Efficient CNNs balance accuracy with parameters, FLOPs, memory, and latency [7, 8]. EfficientNet jointly scales depth, width, and resolution; EfficientNetV2 adds Fused-MBConv and training-aware design; ConvNeXt modernizes convolutional architecture [9]. This thesis uses these architecture families within the proposed hybrid models.

2.1.10 Hybrid Deep Learning Architectures

Hybrid architectures combine complementary representations, such as CNN local bias and transformer context [28]. ABS Hybrid concatenates EfficientNetV2-M and ConvNeXt-Base features; UgvNet-DS fuses DenseNet121 with Swin V2; and UGVNet gates EfficientNetV2-S and ConvNeXt-Tiny features before global attention. TRY NET-8 serves as a compact from-scratch comparator.

2.2 Related Works

2.2.1 CNN and Transfer-Learning Studies

Esteva et al. [3] demonstrated that convolutional neural networks could classify skin lesions using a large collection of clinical images. Their study showed the strong potential of deep

Table 2.2: Published SkinDisNet baseline performance.

Published Baseline	Accuracy	Reported Precision	Reported Recall	Reported F1-Score
EfficientNetB0	86.93%	82.52%	84.71%	83.20%
ResNet50	86.36%	85.06%	86.67%	85.49%
VGG16	86.93%	88.12%	86.76%	87.34%

learning for computer-assisted skin-image analysis. Liu et al. [4] later developed a deep-learning system that used clinical photographs and patient metadata to support the differential diagnosis of skin diseases. These studies established an important foundation for applying CNNs and transfer learning in dermatology. However, they used different datasets, disease categories, image sources, and evaluation methods. Therefore, their reported results cannot be directly compared with the results obtained using the SkinDisNet dataset.

2.2.2 Architecture and Efficiency Context

VGG uses a simple sequence of convolutional layers to learn visual features at different levels [5]. ResNet introduces residual connections that support the training of deeper networks and reduce gradient-related problems [6]. EfficientNet balances network depth, width, and input resolution through compound scaling [7]. ConvNeXt improves conventional CNN design using several ideas found in modern vision architectures, while Swin Transformer learns hierarchical features through window-based self-attention [9, 23, 10]. These architectures provide different forms of local, global, and multi-scale feature extraction. However, larger or more complex models are not always more effective, so they should be compared with compact models under the same dataset, training conditions, and evaluation measures.

2.2.3 Class Imbalance and Evaluation Reliability

Medical-image datasets often contain unequal numbers of samples across disease classes. This imbalance can make a model focus more strongly on common classes while performing poorly on minority classes. Class-balanced loss reduces this problem by assigning greater importance to classes with fewer effective training samples [16]. Overall accuracy alone may hide such weaknesses because it is strongly influenced by majority-class predictions. Macro-averaged and class-wise metrics give each class greater visibility and provide a more balanced view of model performance. Since differences in subgroup representation can also affect the

reliability of dermatology-AI systems [14], this study reports macro-F1, per-class results, confusion patterns, and class support.

2.2.4 Hybrid, Attention, and Explainability Studies

Attention mechanisms help neural networks give more importance to useful features and reduce the effect of less relevant information. Squeeze-and-Excitation networks recalibrate channel features, while CBAM combines channel and spatial attention [21, 22]. Self-attention also supports interaction between distant image regions and helps capture wider visual relationships [20]. CoAtNet demonstrates that convolution and attention can be combined within a single architecture to learn both local and global features [28]. These ideas are relevant to skin disease classification because disease signs may appear as small textures or as patterns spread across a larger skin area. Grad-CAM can help inspect which image regions influence a prediction [29]. However, saliency maps cannot independently prove clinical validity or causal reasoning and must be interpreted carefully [30].

2.2.5 SkinDisNet Context

SkinDisNet is the most directly related benchmark because it contains 1,710 clinical photographs representing six skin disease classes [12]. The published study evaluated EfficientNetB0, ResNet50, and VGG16 on this dataset. Their reported accuracies were 86.93%, 86.36%, and 86.93%, respectively. These results provide a useful reference point for understanding the difficulty of the dataset and the performance of common transfer-learning models. However, they are treated as external reference results because the present study uses a different experimental pipeline, including its own data split, augmentation process, training settings, and evaluation procedure. The present study extends the benchmark to seven models, covering conventional baselines, a compact from-scratch network, and dual-backbone hybrid architectures. Comparisons within this study are therefore based mainly on models evaluated under the same internal experimental conditions.

2.2.6 Research Gap

Existing research on the SkinDisNet benchmark mainly focuses on standard models like EfficientNetB0, ResNet50, and VGG16 [12]. However, prior studies lack a complete comparison

that tests lightweight from-scratch models alongside hybrid multi-backbone designs under a single, controlled system. Important factors such as minority-class performance, strict data leakage controls, and computational efficiency are often missing. This thesis fills these research gaps by evaluating seven distinct models under identical conditions. Additionally, hardware constraints on GPU memory required careful checkpointing and model selection, while keeping earlier test results clearly documented.

Chapter III

Methodology

3.1 Overview of the Experimental Framework

This chapter describes the evaluation of seven deep-learning architectures for six-class SkinDis-Net classification [11, 12]. The framework covers dataset auditing, preprocessing, stratified partitioning, training-only augmentation, class-imbalance handling, model construction, validation-based checkpointing and multi-metric evaluation.

All seven configurations share the same six-class data partition and label order (AD, CD, EC, SC, SD, TC), while model-specific preprocessing, input resolutions, and optimization hyperparameters are documented transparently. Six runs use validation-based model selection and reserve the locked test set for final assessment. The historical UgvNet-DS run is the documented exception because test-set feedback influenced stopping and its metrics are therefore reported only as test-informed development results.



Figure 3.1: Methodology workflow for skin disease classification using four deep learning models.

3.2 Dataset Description and Disease Taxonomy

3.2.1 Dataset Composition

The source dataset consists of 1,710 original clinical photographs collected across 416 patients in hospital dermatology departments. An automated audit was conducted to verify file

integrity, dimensions, color spaces and hash signatures across partitions.

3.2.2 Disease Classes and Per-Class Distribution

The dataset encompasses six prevalent dermatological conditions. Table 3.1 lists the official disease codes and their clinical taxonomic designations, while Table 3.2 details the per-class sample distribution across training, validation, and test partitions. Figure 3.2 provides representative visual examples of the six target categories.

Table 3.1: Target dermatological disease classes and standard codes.

Code	Disease Name
AD	Atopic Dermatitis
CD	Contact Dermatitis
EC	Eczema
SC	Scabies
SD	Seborrheic Dermatitis
TC	Tinea Corporis

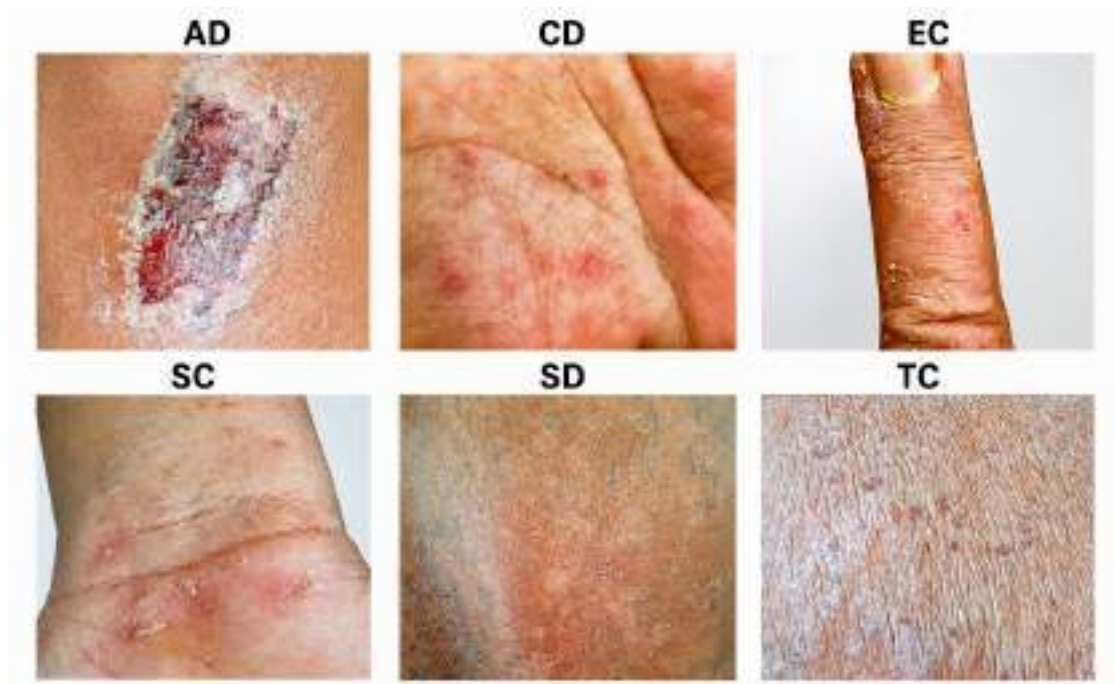


Figure 3.2: Representative clinical image samples from the six target disease categories in SkinDisNet.

3.3 Dataset Partitioning and Training-Only Augmentation

A total of 1,710 original clinical photographs were partitioned at the image level using a 70:15:15 stratified ratio, yielding 1,197 training originals, 256 validation images, and 257 test images.

Augmentation was applied only to the training partition. Random flips, rotations ($\pm 20^\circ$), zoom scaling ($[0.85, 1.15]$), brightness adjustment, and color jittering expanded 1,197 training originals to 9,576 images. Validation and test images remained unchanged. Table 3.2 reports the counts; Figures 3.3 and 3.4 show the split and transformations.

Table 3.2: SkinDisNet class distribution and data partitions.

Code	Disease Category	Orig. Train (70%)	Aug. Train	Val. (15%)	Test (15%)	Total Orig.
AD	Atopic Dermatitis	49	392	11	10	70
CD	Contact Dermatitis	334	2,672	71	72	477
EC	Eczema	326	2,608	70	70	466
SC	Scabies	240	1,920	51	52	343
SD	Seborrheic Dermatitis	55	440	12	12	79
TC	Tinea Corporis	193	1,544	41	41	275
Total		1,197	9,576	256	257	1,710

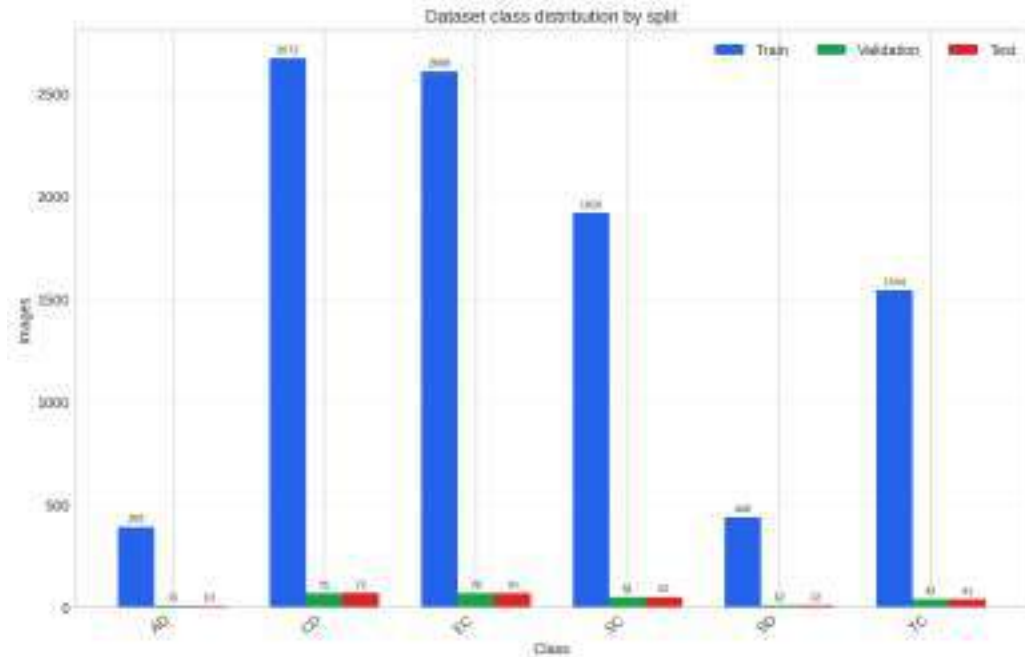


Figure 3.3: Class-wise image count distribution across training, validation, and test partitions.

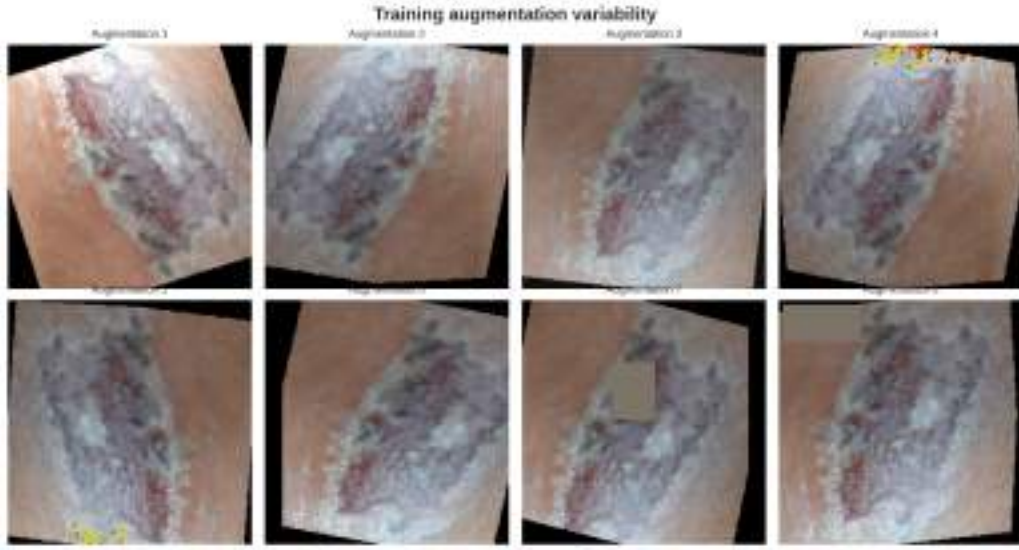


Figure 3.4: Training-only augmentation applied to clinical skin images.

3.4 Class-Imbalance Mitigation Strategies

Because the dataset is imbalanced ($n = 477$ for CD and $n = 70$ for AD), standard cross-entropy can favor majority classes. Class-balanced focal loss was therefore used.

The effective number of samples for class c is [16]:

$$E_{n_c} = \frac{1 - \beta^{n_c}}{1 - \beta} \quad (3.1)$$

where $\beta \in [0, 1)$. The class weight is:

$$W_c = \frac{1 - \beta}{1 - \beta^{n_c}} \quad (3.2)$$

Class-balanced focal loss is:

$$\mathcal{L}_{\text{CB-FL}}(\mathbf{p}, y) = -\frac{1 - \beta}{1 - \beta^{n_y}} (1 - p_t)^\gamma \log(p_t) \quad (3.3)$$

where p_t is the predicted probability of the true class and $\gamma \geq 0$ down-weights well-classified examples.

3.5 Experimental Design and Seven-Model Lineup

The comparison includes three baselines and four proposed models. Table 3.3 summarizes the proposed architectures; Table 3.4 states the shared controls and permitted model-specific variations.

3.5.1 Four Proposed Architectures

1. **ABS Hybrid:** EfficientNetV2-M and ConvNeXt-Base with global pooling and feature concatenation (141.91M parameters).
2. **UgvNet-DS:** DenseNet121 and Swin Transformer V2-Tiny with branch-specific CBAM attention (35.66M parameters).
3. **TRY NET-8:** From-scratch residual MBConv CNN with depthwise convolution, Squeeze-and-Excitation, and stochastic depth (3.49M parameters).
4. **UGVNet:** EfficientNetV2-S and ConvNeXt-Tiny with adaptive gated fusion and 8-head global self-attention (49.46M parameters).

Table 3.3: Architecture and fusion specifications of the proposed models.

Model	Input	Main Components	Fusion & Classifier	Parameters
ABS Hybrid	224×224	EfficientNetV2-M + ConvNeXt-Base; pretrained; frozen backbones	GAP; concatenation; BN; 512-unit head	141,911,354
UgvNet-DS	300×300 ; Swin 256×256	DenseNet121 + Swin V2 Tiny; CBAM; partial fine-tuning	Branch GAP; concatenation; 512-unit head	Total params: 35,658,100 Trainable: 18,683,706 (52.4%)
TRY NET-8	256×256	12 residual MB-Conv blocks; SE; stochastic depth	Dual GAP/GMP; 384-unit head	3,488,202
UGVNet	224×224	EfficientNetV2-S + ConvNeXt-Tiny; adaptive gate; 8-head attention	Weighted fusion; GAP; linear head	49,458,872 total; 1,462,792 trainable

3.5.2 Three Baseline Architectures

1. **Conventional CNN:** Custom four-stage convolutional network trained from scratch.
2. **VGG16:** Sequential CNN pretrained on ImageNet [5].
3. **ResNet50:** Residual network with bottleneck blocks and identity shortcuts [6].

Table 3.4: Shared experimental controls and model-specific variations.

Factor	Experimental Rule
Task and labels	Six-way image classification with fixed code order: AD, CD, EC, SC, SD, TC.
Dataset partitions	Fixed 9,576 training, 256 validation, and 257 test images without cross-partition movement.
Training information	Class balancing and augmentation decisions derived strictly from training data.
Model selection	Validation macro-F1 serves as primary checkpoint criterion; accuracy/loss serve as tie-breakers.
Test access	Evaluated once after model architecture and inference policies are locked, except for the historical UgvNet-DS run, which is marked as test-informed.
Evaluation script	Unified evaluation script applying identical metric implementations to saved predictions.
Permitted variation	Resolution, batch size, optimizer, learning rate, loss, and freezing vary by architecture.
Test-time augmentation	Single-view inference reported as primary confirmatory result.
Resource reporting	Total/trainable parameters, input dimensions, training duration, and memory footprint documented.

3.6 Proposed Model Architectures and Mechanics

3.6.1 ABS Hybrid

ABS Hybrid combines EfficientNetV2-M and ConvNeXt-Base in a parallel dual-branch topology. Both backbones receive 224×224 images and extract spatial feature maps. Global Average Pooling (GAP) reduces each branch to a 1D feature vector: $\mathbf{f}_{\text{eff}} \in \mathbb{R}^{1280}$ and $\mathbf{f}_{\text{conv}} \in \mathbb{R}^{1024}$. The vectors are concatenated into $\mathbf{f}_{\text{fused}} \in \mathbb{R}^{2304}$, normalized via Batch Normalization, regularized with dropout ($p = 0.40$), and projected through a 512-unit dense layer to the six-class softmax classifier. Figures 3.5 and 3.6 show the parallel backbone and the fused classification head, respectively.

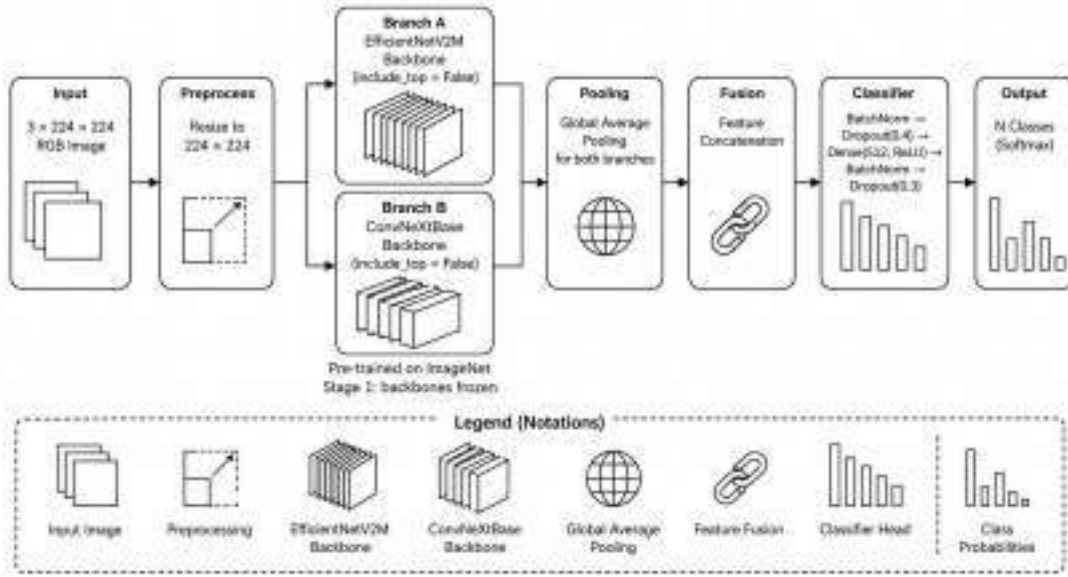


Figure 3.5: ABS Hybrid dual-backbone architecture.

Proposed Hybrid EfficientNetV2M–ConvNeXtBase Architecture

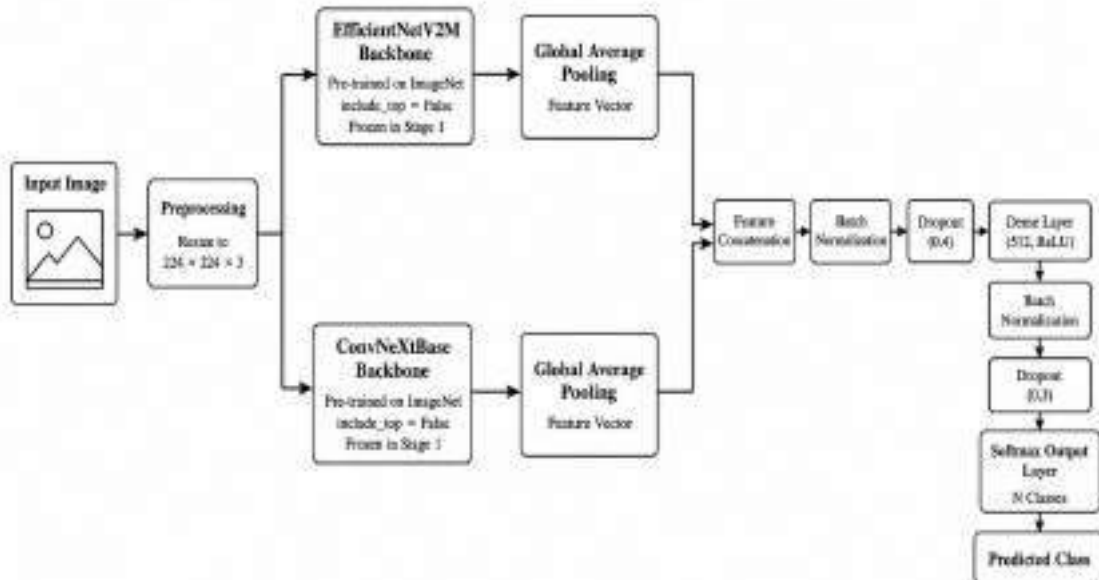


Figure 3.6: ABS Hybrid fusion and classification head.

3.6.2 UgvNet-DS

UgvNet-DS integrates DenseNet121 and Swin Transformer V2-Tiny. DenseNet121 processes 300×300 images, while Swin Transformer V2 processes 256×256 patches. Spatial feature maps from each branch pass through Convolutional Block Attention Modules (CBAM) to refine spatial and channel salience before global pooling, concatenation, and classification through a 512-unit dense head, as summarized in Figure 3.7.

3.6.3 TRY NET-8

TRY NET-8 is a custom from-scratch Attentive Residual MBConv CNN engineered for high parameter efficiency (3.49M parameters). It comprises a 32-channel initial stem followed by four residual stages (48, 96, 160, and 256 channels) containing 2, 3, 4, and 3 residual MBConv blocks, respectively. Each block utilizes 1×1 expansion, 5×5 depthwise convolution, Squeeze-and-Excitation channel attention ($r = 4$), and stochastic depth drop-path regularization. The final 384-channel feature map undergoes dual pooling (GAP + GMP) before classification, as shown in Figure 3.8.

3.6.4 UGVNet

UGVNet combines EfficientNetV2-S and ConvNeXt-Tiny with Adaptive Gated Feature Fusion and 8-head Global Self-Attention. Both branches extract feature maps $\mathbf{F}_{\text{eff}} \in \mathbb{R}^{H \times W \times C_1}$ and $\mathbf{F}_{\text{conv}} \in \mathbb{R}^{H \times W \times C_2}$. 1×1 projections align both representations to a common dimension $C = 256$.

Adaptive Gated Fusion: The projected maps $\mathbf{F}'_{\text{eff}}$ and $\mathbf{F}'_{\text{conv}}$ are concatenated and passed through a 1×1 convolution with softmax activation to generate spatial gate maps $\mathbf{G}_1, \mathbf{G}_2 \in \mathbb{R}^{H \times W \times 1}$ ($\mathbf{G}_1 + \mathbf{G}_2 = 1$):

$$\mathbf{F}_{\text{gated}} = \mathbf{G}_1 \odot \mathbf{F}'_{\text{eff}} + \mathbf{G}_2 \odot \mathbf{F}'_{\text{conv}} \quad (3.4)$$

Global Self-Attention Refinement: The gated feature map passes through depthwise positional encoding, 8-head multi-head self-attention (MHSA), Layer Normalization, and a feed-forward network with residual connections:

$$\mathbf{F}_{\text{attn}} = \text{MHSA}(\mathbf{F}_{\text{gated}}) + \mathbf{F}_{\text{gated}} \quad (3.5)$$

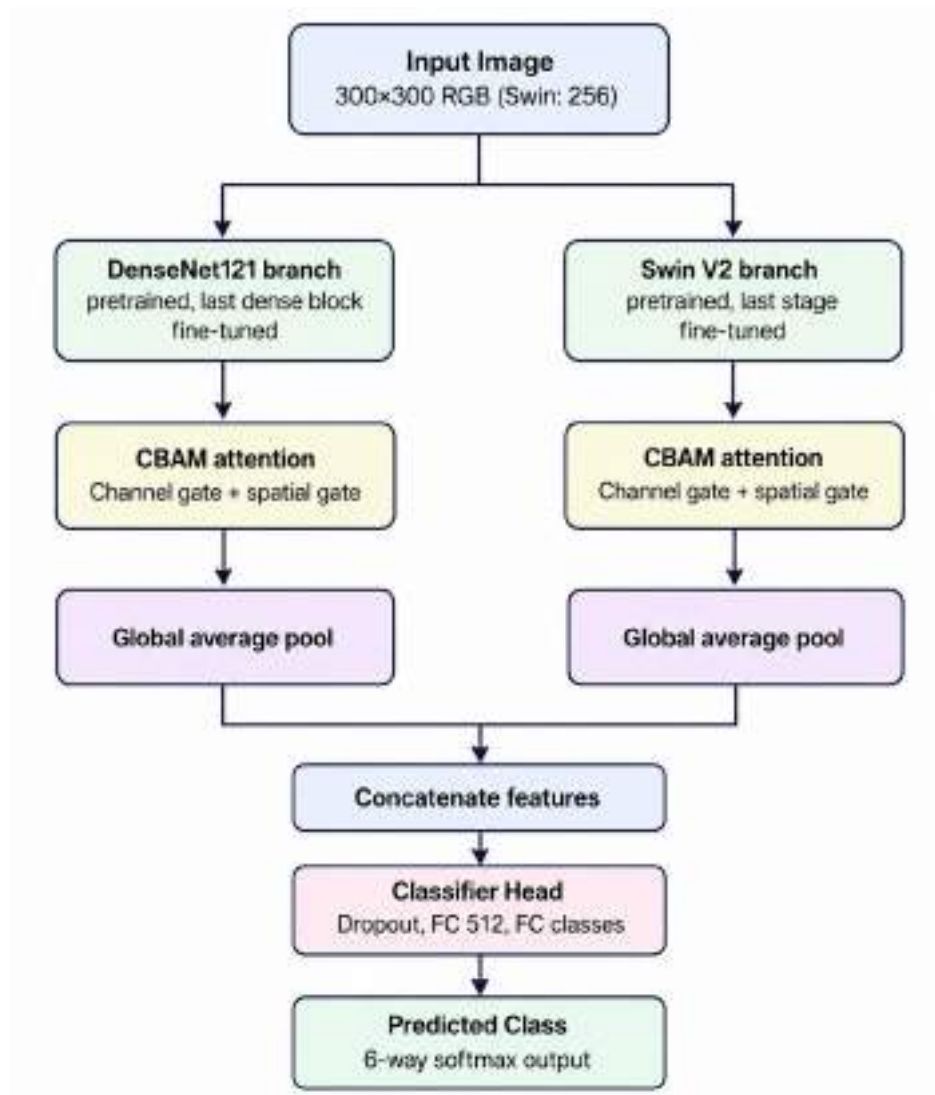


Figure 3.7: UgyNet-DS CNN-Transformer architecture with CBAM.

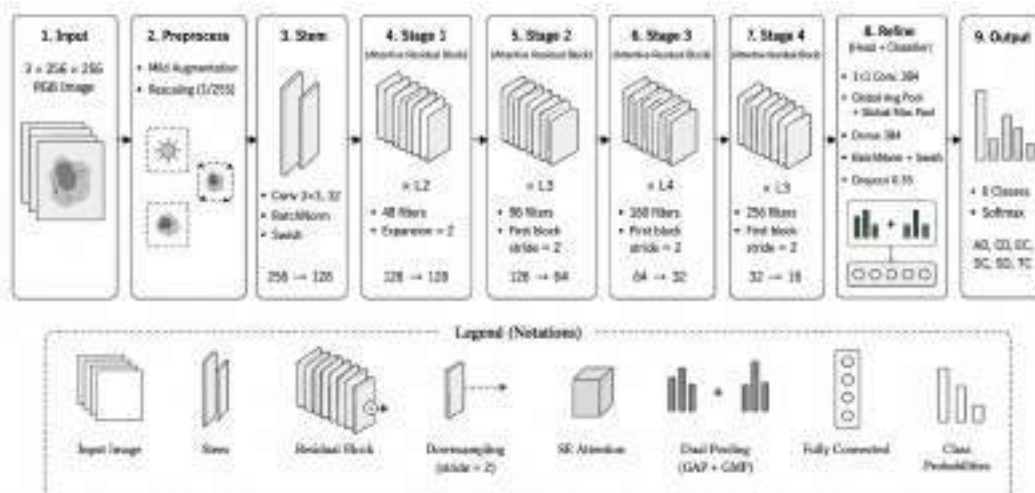


Figure 3.8: TRY NET-8 attentive residual CNN architecture.

The refined map is globally pooled, regularized with 0.20 dropout, and classified via a linear head. Figure 3.9 presents the end-to-end topology, and Figure 3.10 details the gated fusion and attention mechanism.

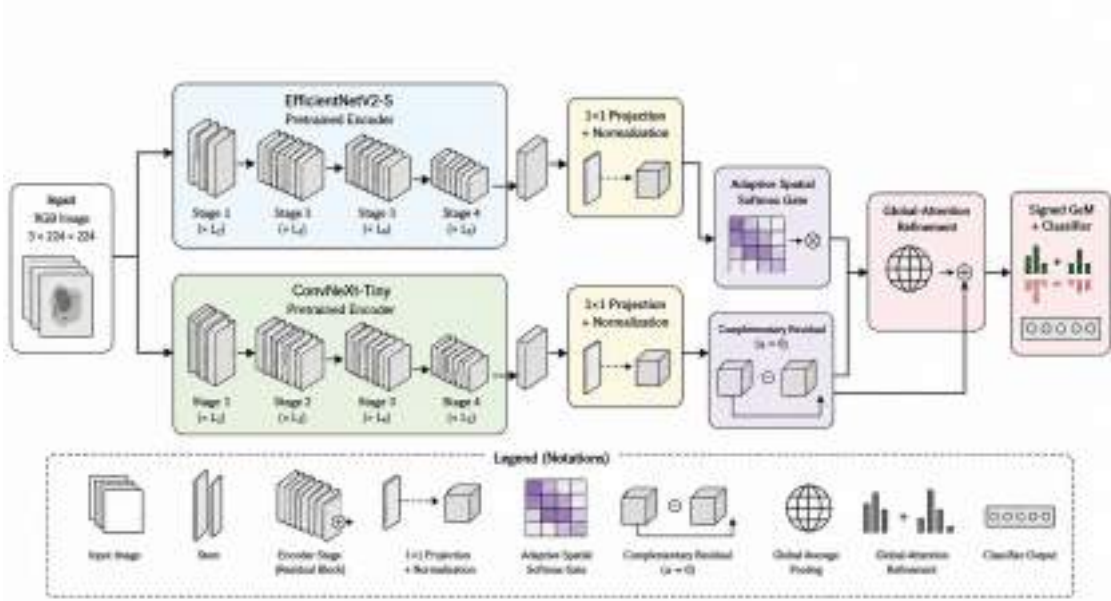


Figure 3.9: UGVNet dual-backbone architecture.

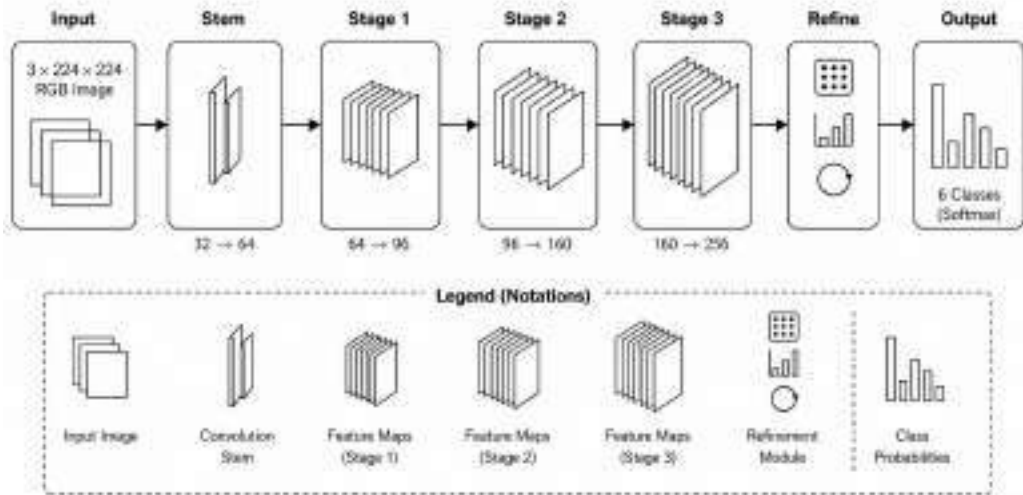


Figure 3.10: UGVNet adaptive gated fusion and global self-attention.

3.7 Evaluation Metrics and Mathematical Formulations

To ensure rigorous and multi-faceted evaluation beyond aggregate accuracy, performance is quantified using precision, recall, F1-score, macro-averaged metrics, balanced accuracy, and

Expected Calibration Error:

$$\text{Accuracy} = \frac{\sum_{c=1}^C \text{TP}_c}{N} \quad (3.6)$$

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (3.7)$$

$$\text{F1}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (3.8)$$

$$\text{Macro Precision} = \frac{1}{C} \sum_{c=1}^C \text{Precision}_c, \quad \text{Macro Recall} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c \quad (3.9)$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c \quad (3.10)$$

where TP_c , FP_c , FN_c denote true positives, false positives, and false negatives for class c , $C = 6$ is the number of diagnostic categories, and $N = 257$ is the total number of test samples.

Chapter IV

Results and Discussion

4.1 Experimental Results Overview

This chapter compares three baselines (Conventional CNN, VGG16, and ResNet50) with four proposed models (ABS Hybrid, UgvNet-DS, TRY NET-8, and UGVNet). Evaluation covers convergence, test performance, class-wise behavior, error patterns, qualitative cases, latent embeddings, and model complexity.

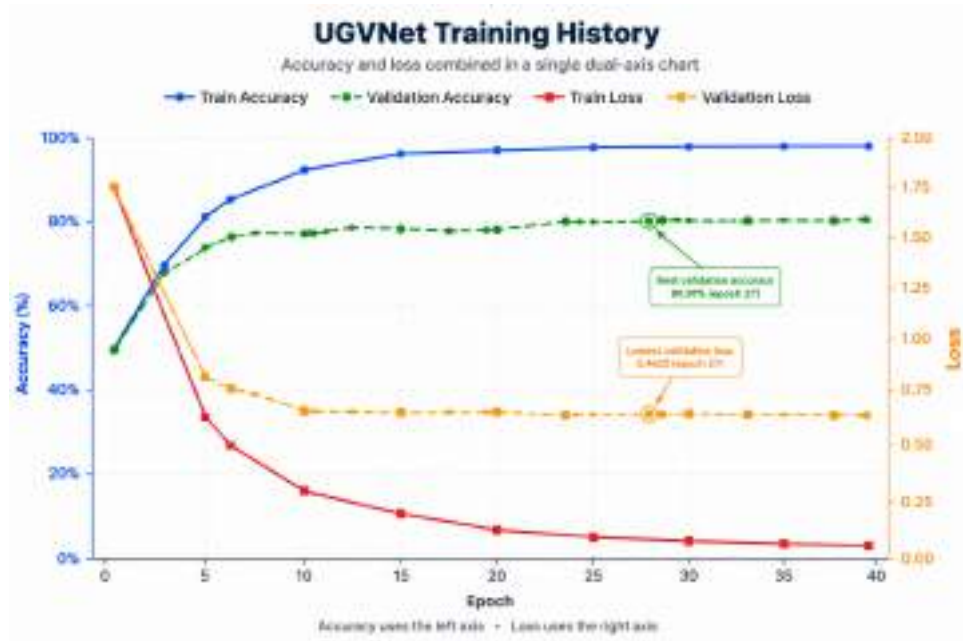
4.2 Training and Validation Performance

Table 4.1 summarizes model convergence and train–validation separation.

Table 4.1: Training and validation performance by model

Model	Final Train Acc.	Best/Final Val. Acc.	Main Observation
Conventional CNN	67.30%	64.84%	Slow learning with modest train–validation gap
VGG16	91.32%	76.17%	Strong training gain; noticeable generalization gap
ResNet50	97.26%	84.77%	Strong fitting with validation saturation
ABS Hybrid	95.39%	87.11% (epoch 9)	Optimal checkpoint at epoch 9 (8.02 pp gap)
UgvNet-DS	99.69%	85.55%	Strong fitting; historical run test-informed
TRY NET-8	$\approx 99.69\%$	89.45%	Strong from-scratch learning; separation in late epochs
UGVNet	99.98%	90.23%	Highest validation accuracy; later-stage overfitting

The curves in Figure 4.1 show rapid fitting by the pretrained hybrids and later train–validation separation. UgvNet-DS reached 89.45% validation accuracy after 25 epochs, ABS Hybrid peaked at 87.11%, and TRY NET-8 reached 89.45% despite using only 3.49M parameters.



(a) UGVNet training and validation curves.

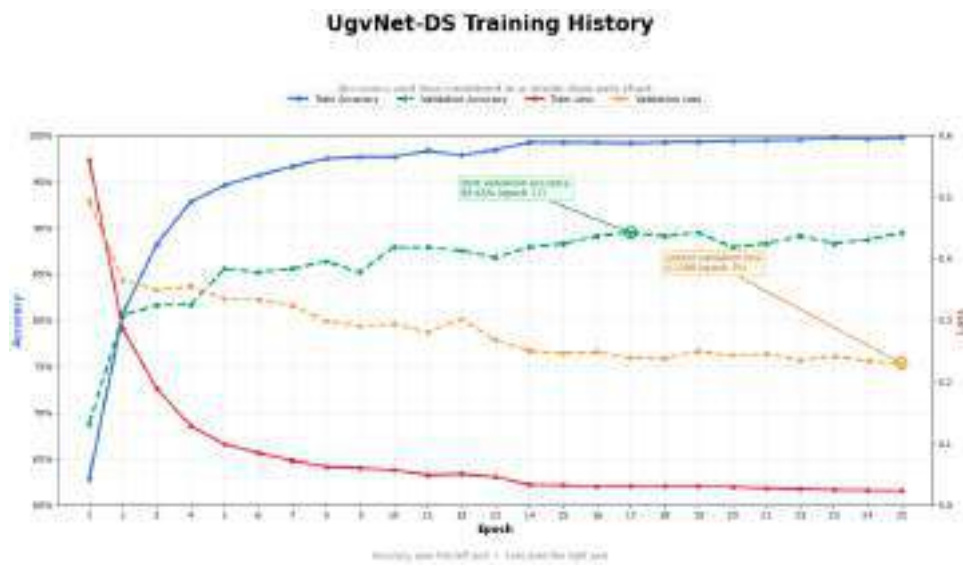


(b) ABS Hybrid training and validation curves

Figure 4.1: Training and validation curves for the proposed models.



(c) TRY NET-8 training and validation curves.



(d) UgvNet-DS training and validation curves.

Figure 4.1: Training and validation curves for the proposed models (continued).

4.3 Final Test-Set Performance and Overall Model Rankings

Table 4.2 reports results on the locked 257-image test set. UGVNet ranked first (91.83% accuracy; 0.9250 macro-F1), followed by ABS Hybrid (90.66%; 0.8877). TRY NET-8 matched ResNet50 at 85.21% accuracy with 3.49M parameters.

Table 4.2: Final test performance and model ranking.

Rank	Model	Group	Test Acc.	Macro Prec.	Macro Rec.	Macro-F1	Evidence Status
1	UGVNet	Proposed	91.83%	0.9222	0.9295	0.9250	Validation-selected
2	ABS Hybrid	Proposed	90.66%	0.9169	0.8776	0.8877	Validation-selected
3	UgvNet-DS	Proposed	90.27%	0.8989	0.8858	0.8919	Test-informed result
4	ResNet50	Baseline	85.21%	0.8800	0.8200	0.8400	Validation-selected
5	TRY NET-8	Proposed	85.21%	0.8407	0.8137	0.8170	From-scratch validation-selected
6	VGG16	Baseline	80.00%	0.7700	0.7400	0.7500	Transfer-learning reference
7	CNN	Baseline	63.00%	0.5100	0.5400	0.5200	From-scratch baseline

Historical UgvNet-DS run is test-informed due to epoch-level test-set monitoring.

4.4 Class-Wise Performance and Error Analysis

4.4.1 Class-Wise Precision, Recall, and F1 Profiles

Tables 4.3, 4.4, and 4.5 detail the class-specific diagnostic performance across all six disease classes.

Table 4.3: Per-class precision by model.

Model	AD	CD	EC	SC	SD	TC
Conventional CNN	0.5000	0.7471	0.8178	0.8321	0.7898	0.8090
VGG16	0.5600	0.7600	0.8300	0.8500	0.8000	0.8100
ResNet50	0.6759	0.8765	0.8678	0.8687	0.8231	0.9645
ABS Hybrid	1.0000	0.8947	0.9104	0.8889	0.8571	0.9500
UgvNet-DS	0.8000	0.8846	0.9077	0.9231	0.9091	0.9268
TRY NET-8	0.8333	0.7952	0.9048	0.8936	0.7333	0.8837
UGVNet	0.9091	0.8961	0.8955	0.9423	0.9167	0.9737

Table 4.4: Per-class recall by model.

Model	AD	CD	EC	SC	SD	TC
Conventional CNN	0.5100	0.8132	0.7713	0.7621	0.7788	0.7272
VGG16	0.5000	0.8333	0.8143	0.7885	0.6667	0.8293
ResNet50	0.49333	0.8222	0.7876	0.8078	0.7998	0.9089
ABS Hybrid	0.6000	0.9444	0.8714	0.9231	1.0000	0.9268
UgvNet-DS	0.8000	0.9583	0.8429	0.9231	0.8333	0.9268
TRY NET-8	0.5000	0.9167	0.8143	0.8077	0.9167	0.9268
UGVNet	1.0000	0.9583	0.8571	0.9423	0.9167	0.9024

Table 4.5: Per-class F1-score by model.

Model	AD	CD	EC	SC	SD	TC
Conventional CNN	0.6000	0.7865	0.7989	0.7656	0.8198	0.9331
VGG16	0.5300	0.7900	0.8200	0.8200	0.7300	0.8200
ResNet50	0.7500	0.8500	0.8500	0.8800	0.8800	0.8200
ABS Hybrid	0.7500	0.9189	0.8905	0.9057	0.9231	0.9383
UgvNet-DS	0.8000	0.9200	0.8741	0.9231	0.8696	0.9268
TRY NET-8	0.6250	0.8516	0.8571	0.8485	0.8148	0.9048
UGVNet	0.9524	0.9262	0.8759	0.9423	0.9167	0.9367

4.4.2 Minority-Class Sensitivity and Class Imbalance Impact

Table 4.6 evaluates performance on minority categories. In particular, underrepresented classes such as Atopic Dermatitis (10 test samples) and Seborrheic Dermatitis (12 test samples) pose substantial sensitivity challenges for conventional baseline architectures due to severe training class imbalance.

4.4.3 Confusion Patterns and Cross-Model Error Dynamics

Tables 4.7 and 4.8 document specific confusion pairs and recurring morphological overlap patterns. The primary confusion axis across models occurred between Eczema (EC) and Contact Dermatitis (CD), reflecting their shared erythematous plaque morphology.

Table 4.7 identifies EC–CD as the main error axis. ABS Hybrid misclassified 24 of 257 samples; EC→CD and EC→SC were its most frequent errors (four each), while AD errors were dispersed across categories.

Figures 4.2–4.4 support the tabulated error analysis. UgvNet-DS shows limited bidirectional EC–CD confusion; TRY NET-8 has stronger SC→CD and EC→CD errors; and UGVNet

Table 4.6: AD and SD sensitivity and F1-scores.

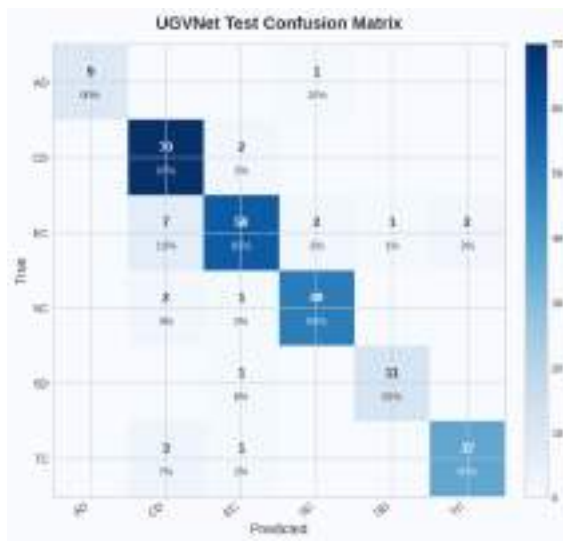
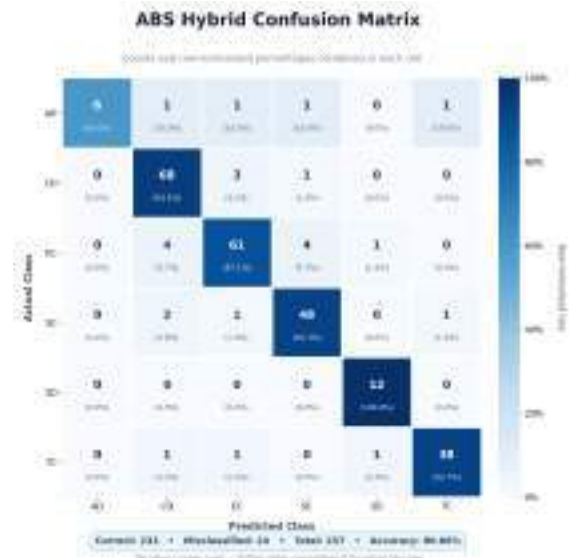
Model	AD Supp.	AD Rec.	AD F1	SD Rec.	SD F1	Observation
Conventional CNN	10	0.5100	0.5422	0.6789	—	Complete AD recognition failure
VGG16	10	0.5000	0.5300	0.6667	0.7300	Substantial minority-class degradation
ResNet50	10	0.6578	0.7500	0.6897	0.8800	Strongest base-line balance
ABS Hybrid	10	0.6000	0.7500	1.0000	0.9231	Perfect SD recall; moderate AD sensitivity
UgvNet-DS	10	0.8000	0.8000	0.8333	0.8333	Strong AD result (test-informed)
TRY NET-8	10	0.5000	0.6250	0.9167	0.8148	AD sensitivity principal bottleneck
UGVNet	10	1.0000	0.9524	0.9167	0.9167	Perfect 10/10 AD recall; optimal balance

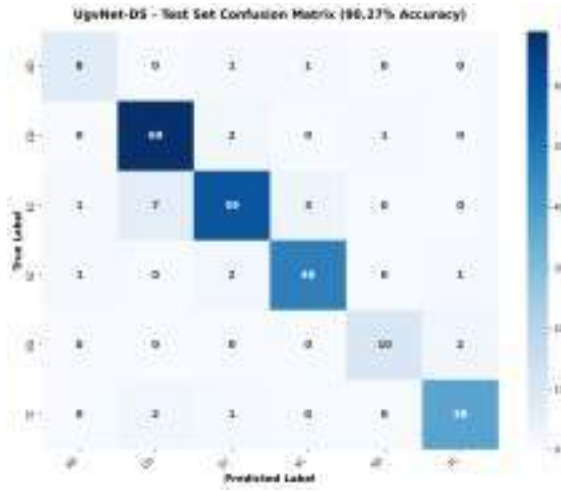
Table 4.7: Main misclassification pairs by proposed model.

Model	True	Pred.	Count	Morphological Context
ABS Hybrid	EC	CD	4	Primary EC confusion direction
ABS Hybrid	EC	SC	4	Papular and scaling overlap
ABS Hybrid	CD	EC	3	Bidirectional CD–EC ambiguity
ABS Hybrid	SC	CD	2	Secondary erythema confusion
ABS Hybrid	AD	SC / TC	2	Minority class misclassification
ABS Hybrid	SD	Other	0	Perfect classification (12/12)
UGVNet	EC	CD	6	Principal UGVNet error mode
UGVNet	CD	EC	3	Reverse CD–EC ambiguity
UGVNet	TC	CD	2	Annular erythema mimicry
UGVNet	SC	EC	2	Secondary scaling confusion
UGVNet	AD	Other	0	Perfect classification (10/10)
TRY NET-8	EC	CD	7	Major EC error direction
TRY NET-8	SC	CD	7	Pronounced CD bias on scabies
TRY NET-8	AD	Other	5	5/10 AD samples misclassified

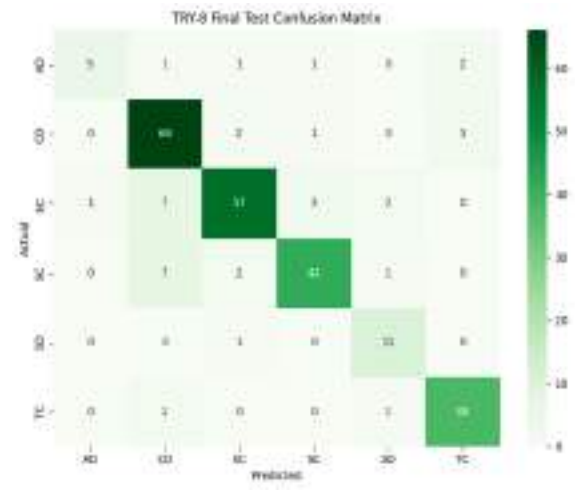
Table 4.8: Recurring error patterns and interpretations.

Pattern	Models Affected	Evidence	Interpretation
AD recognition failure	Conv. CNN, TRY-8	CNN AD F1=0.00; TRY-8 F1=0.625	AD most sensitive to severe class imbalance
EC → CD confusion	ABS Hybrid, UGVNet, TRY-8	ABS: 4; UGVNet: 6; TRY-8: 7	Morphological boundary between EC and CD inherently diffuse
CD → EC confusion	ABS Hybrid, UGVNet	ABS: 3; UGVNet: 3	Bidirectional visual ambiguity in erythematous plaques
SC → CD confusion	TRY NET-8	7 SC images predicted as CD	Compact receptive fields struggle with diffuse scabietic lesions
Robust SD recognition	ABS Hybrid, UGVNet, TRY-8	Recall = 1.00, 0.9167, 0.9167	Greasy yellowish scale provides distinct visual signature
Uniform class balance	UGVNet	Every class F1 ≥ 0.8759	Adaptive gating prevents minority-class suppression

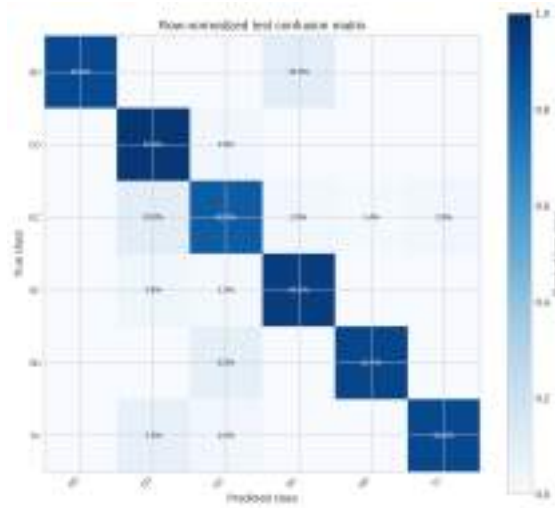
**(a)** UGVNet confusion matrix (counts).**(b)** ABS Hybrid confusion matrix (counts).**Figure 4.2:** Test-set confusion matrices for UGVNet and ABS Hybrid.



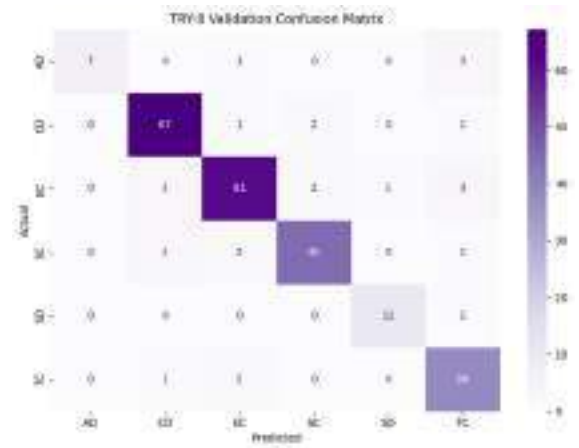
(a) UgvNet-DS test confusion matrix.



(b) TRY NET-8 test confusion matrix.

Figure 4.3: Test confusion matrices for UgvNet-DS and TRY NET-8.

(a) UGVNet normalized confusion matrix.



(b) TRY NET-8 validation confusion matrix.

Figure 4.4: UGVNet normalized test matrix and TRY NET-8 validation matrix.

maintains high normalized rates across all classes. The TRY NET-8 validation matrix is shown separately from its final test matrix.

4.4.4 Qualitative Clinical Case Studies of Diagnostic Misclassification

Figures 4.5 and 4.6 show representative ABS Hybrid and UGVNet test errors.



Figure 4.5: ABS Hybrid test misclassifications with labels and confidence scores.

Dermatological Interpretation of Failure Modes:



Figure 4.6: UGVNet high-confidence test errors.

1. **Eczema vs. Scabies:** Lesions with prominent excoriation papules and secondary crusting on flexural skin frequently trigger high-confidence predictions for Scabies (e.g., reaching 99.7% and 94.8% confidence) due to textural mimicry of scabietic burrows and inflammatory nodules.
2. **Atopic Dermatitis vs. Tinea Corporis:** Erythematous patches with active peripheral desquamation and central clearing closely replicate the classic annular ring morphology of dermatophytosis (Tinea Corporis), leading to high-certainty mispredictions.
3. **Eczema vs. Contact Dermatitis:** Acute ill-defined erythema with vesicular micro-edema shares nearly identical epidermal histopathology between acute eczema and allergic contact dermatitis, making purely image-based discrimination without clinical patch testing inherently challenging.
4. **Minority-Class Dispersion:** The severely underrepresented Atopic Dermatitis (AD) and Seborrheic Dermatitis (SD) classes remain particularly vulnerable to high-confidence misclassifications into majority inflammatory categories when overlapping erythema

patterns occur.

4.5 Latent Feature Space Embeddings and Manifold Separation

To inspect whether hybrid deep networks learn semantically meaningful representations that separate disease categories before the final linear softmax layer, high-dimensional latent vectors extracted by the penultimate fusion layer were visualized using 3D embedding projections.

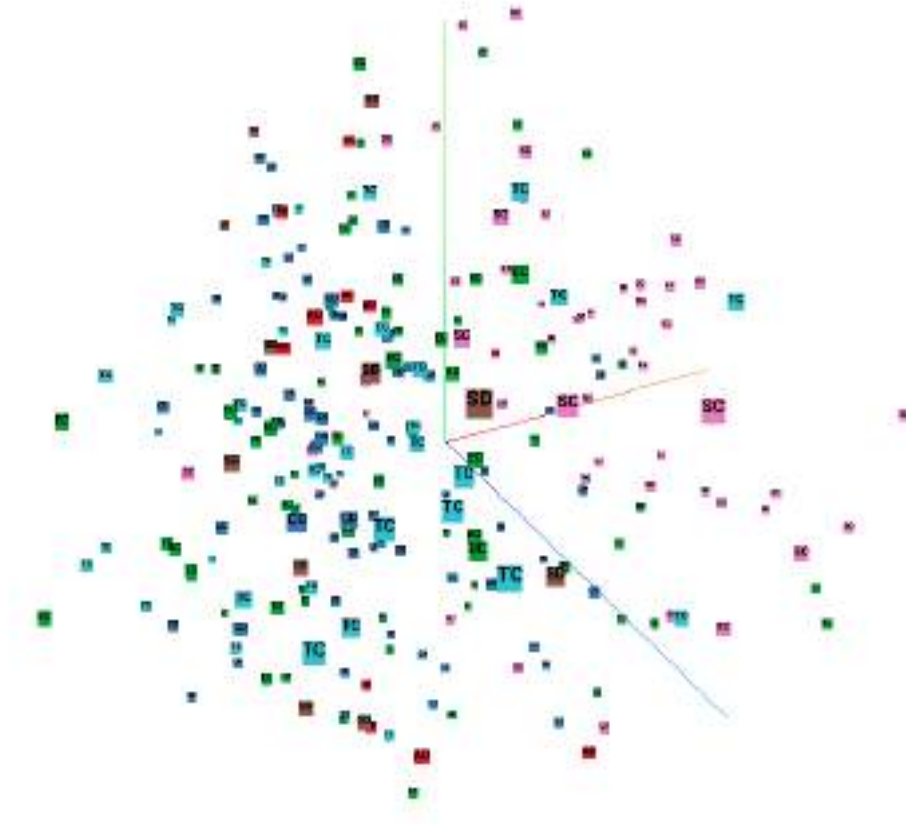


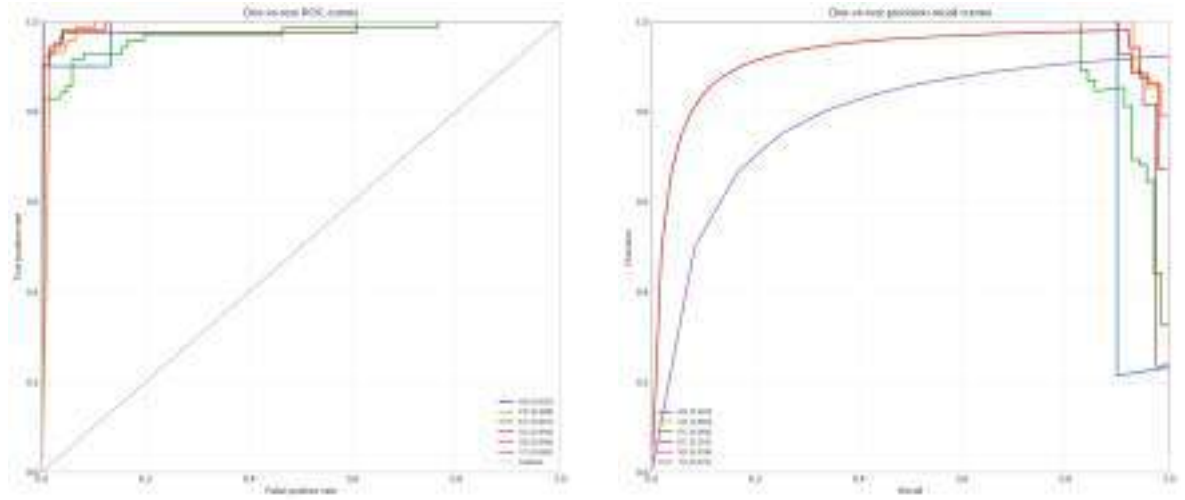
Figure 4.7: 3D projection of UgvNet-DS latent features.

Figure 4.7 shows compact SD and TC clusters, while the proximity of EC and CD is consistent with their confusion-matrix overlap. Because the display is a 3D projection, it is treated as qualitative rather than proof of separability.

4.6 Multi-Class ROC and Calibration Profiling

Figures 4.8, 4.9, and 4.10 display multi-class ROC trajectories, Precision–Recall curves, reliability diagrams, and prediction confidence distributions. Specifically, Figures 4.8a and 4.8b

present UGVNet discrimination curves, while Figures 4.10a and 4.10b present its calibration and confidence profiles.



(a) UGVNet Multi-Class ROC Curves.

(b) UGVNet Precision-Recall Curves.

Figure 4.8: Multi-class ROC and Precision-Recall trajectories for the UGVNet architecture.

4.7 Proposed Models vs. Baseline Models

Table 4.9 compares the proposed models against the highest reported baseline in the published SkinDisNet literature (86.93% [12]). UGVNet achieved a +4.90 percentage point gain, and ABS Hybrid achieved a +3.73 percentage point gain. Table 4.10 expresses the corresponding changes relative to the implemented ResNet50 baseline, while Table 4.11 compares accuracy with macro-F1 to expose class-balance effects.

Table 4.9: Proposed architectures compared with the published SkinDisNet [12] benchmark.

Model	Test Accuracy	Published Best Baseline	Difference	Evidence Status
TRY NET-8	85.21%	86.93%	−1.72 pp	Validation-selected recorded result
ABS Hybrid	90.66%	86.93%	+3.73 pp	Validation-selected recorded result
UgvNet-DS	90.27%	86.93%	+3.34 pp	Test-informed development result
UGVNet	91.83%	86.93%	+4.90 pp	Validation-selected recorded result

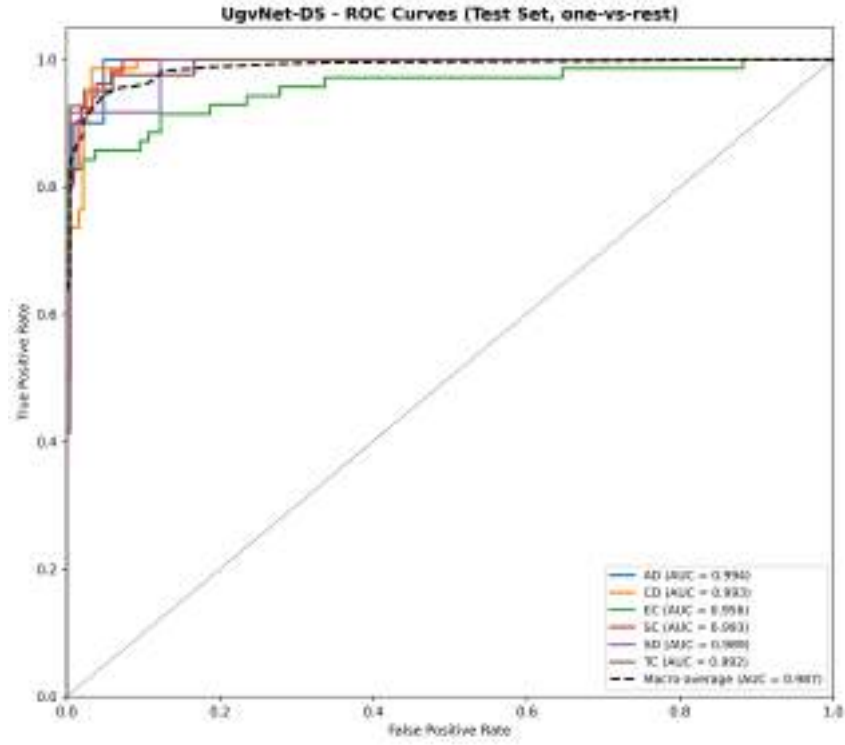
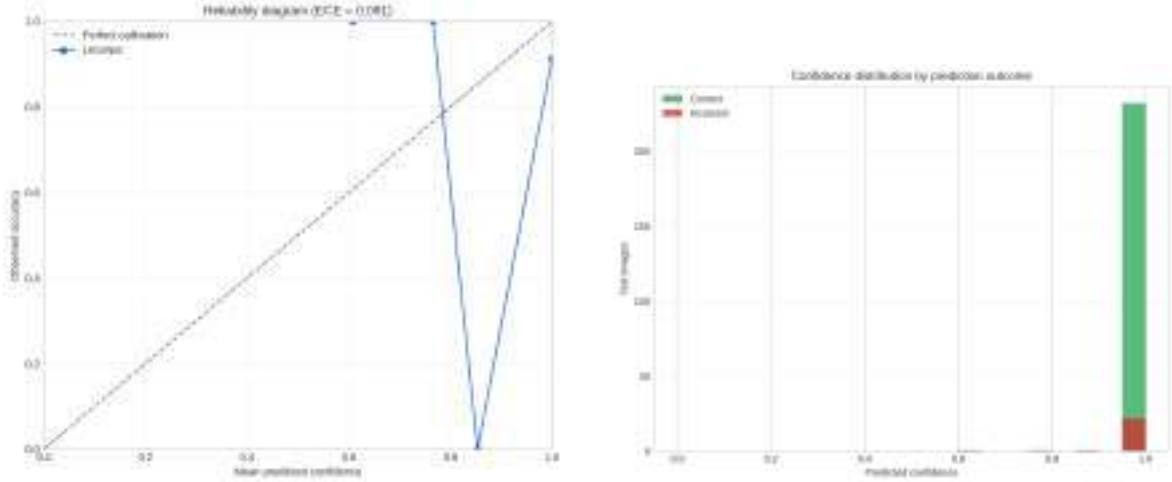


Figure 4.9: UgvNet-DS multi-class ROC curves.



(a) Reliability diagram and calibration.

(b) Confidence distribution.

Figure 4.10: Calibration analysis and prediction confidence profiling for UGVNet.

Table 4.10: Proposed-model performance gains relative to the implemented ResNet50 baseline.

Proposed Model	Δ Accuracy	Δ Macro Prec.	Δ Macro Rec.	Δ Macro-F1
ABS Hybrid	+5.45 pp	+0.0369	+0.0576	+0.0477
UgvNet-DS	+5.06 pp	+0.0189	+0.0658	+0.0519
TRY NET-8	0.00 pp	-0.0393	-0.0063	-0.0230
UGVNet	+6.62 pp	+0.0422	+0.1095	+0.0850

Table 4.11: Accuracy versus macro-F1 discrepancy analysis (class-balance gap).

Model	Accuracy	Macro-F1	Gap (Δ)	Interpretation
Conventional CNN	63.00%	52.00%	11.00 pp	Severe bias toward majority classes
VGG16	80.00%	75.00%	5.00 pp	Moderate class-balance degradation
ResNet50	85.21%	84.00%	1.21 pp	Strong baseline class balance
ABS Hybrid	90.66%	88.77%	1.89 pp	Strong accuracy; AD lowers macro score
UgvNet-DS	90.27%	89.19%	1.08 pp	Strong recorded balance (test-informed)
TRY NET-8	85.21%	81.70%	3.51 pp	Strong overall accuracy; lower AD sensitivity
UGVNet	91.83%	92.50%	-0.67 pp	Superior class balance across all 6 classes

4.8 Qualitative Sample Predictions

Figures 4.11 and 4.12 showcase representative multi-class test inferences across true positive predictions and challenging diagnostic failure cases evaluated on the proposed UgvNet-DS hybrid architecture and the compact from-scratch TRY NET-8 model.

The examples in Figures 4.11 and 4.12 are consistent with the aggregate results: UgvNet-DS usually separates the displayed classes, whereas TRY NET-8 more often confuses CD or AD with EC. These examples are illustrative and do not establish calibration or clinical generalization.

4.9 Model Complexity and Efficiency Comparison

Training strategies, trainable counts, and input resolutions are reported in Table 3.3. Table 4.12 therefore focuses on the nonredundant performance–memory trade-off.

4.10 Discussion

4.10.1 Minority-Class Recognition

UGVNet recalled all 10 AD test cases and maintained the highest macro-F1. This result is consistent with improved minority-class recognition, but the small AD sample and lack of



Figure 4.11: UgvNet-DS test predictions and confidence scores.



Figure 4.12: TRY NET-8 test misclassifications.

Table 4.12: Performance and model-complexity comparison.

Model	Total Params	Weight Mem. (FP32)	Accuracy	Macro-F1	Efficiency Interpretation
TRY NET-8	3.49M	13.95 MB	85.21%	0.8170	Optimal edge-device efficiency
UGVNet	49.46M	197.84 MB	91.83%	0.9250	Best diagnostic power with moderate footprint
UgvNet-DS	35.66M	225.53 MB	90.27	0.8919	Strong development result; clean rerun needed
ABS Hybrid	141.91M	567.65 MB	90.66%	0.8877	High accuracy; largest memory requirements

ablation experiments prevent attributing the gain specifically to gated fusion or attention.

4.10.2 Generalization and Overfitting Behavior

Pretrained dual-backbone models converged rapidly but required aggressive regularization (dropout, LayerNorm, focal loss) to avoid validation saturation. In contrast, TRY NET-8 demonstrated that a compact from-scratch architecture with stochastic depth and Squeeze-and-Excitation attention can match large 23.6M-parameter baselines like ResNet50 on this task.

4.10.3 Performance-to-Efficiency Interpretation

UGVNet achieved the highest test accuracy (91.83%) and macro-F1 (0.9250), but its deployment suitability requires external validation and latency testing. TRY NET-8 uses 13.95 MB of FP32 weights and reached 85.21% accuracy, making it the more compact candidate for future edge-device evaluation.

4.11 Limitations

Several methodological constraints must be highlighted:

1. **Single-Dataset Scope:** Findings are established on SkinDisNet and cannot be generalized across patient populations or imaging modalities without external validation.

2. **Historical UgVNet-DS Qualification:** The UgVNet-DS run utilized test feedback for monitoring; hence its metrics are treated as development findings rather than independent confirmatory estimates.
3. **Absence of Multi-Seed Variance Bounds:** Experiments reflect single controlled runs due to cloud GPU quotas; repeated multi-seed experiments are necessary to establish formal confidence intervals.
4. **Class Imbalance:** Some disease classes contain significantly fewer images than others, which may reduce the reliability of the model's performance on minority classes.
5. **Single Dataset :** The models were evaluated only on the SkinDisNet dataset, so their performance on other hospitals, patient populations, or real-world datasets remains uncertain.

Chapter V

Conclusion and Future Work

5.1 Conclusion

This study empirically compared seven deep-learning configurations for six-class skin disease classification on SkinDisNet. Three established baselines (Conventional CNN, VGG16, and ResNet50) and four proposed architectures (ABS Hybrid, UgvNet-DS, TRY NET-8, and UGVNet) were assessed through accuracy, macro-averaged metrics, balanced accuracy, class sensitivity, error patterns, and model complexity.

ResNet50 was the strongest baseline, achieving 85.21% accuracy and 0.8400 macro-F1. The flagship UGVNet, combining EfficientNetV2-S and ConvNeXt-Tiny with gated fusion and 8-head self-attention, achieved the best overall performance: 91.83% accuracy and 0.9250 macro-F1. It exceeded the highest published SkinDisNet baseline (86.93% [12]) by 4.90 percentage points and recalled all 10 Atopic Dermatitis cases. ABS Hybrid achieved 90.66% accuracy and 0.8877 macro-F1, a 3.73-point improvement over the published baseline.

For resource-constrained deployment, the from-scratch TRY NET-8 matched the 23.6M-parameter ResNet50 at 85.21% accuracy using only 3.49 million parameters (≈ 13.95 MB), offering a strong compactness–performance trade-off. UgvNet-DS recorded 90.27% accuracy and 0.8919 macro-F1, but remains a test-informed development result.

These dataset-specific findings do not establish clinical superiority or deployment readiness; they show how multi-backbone fusion, attention gating, and compact residual modeling address visual mimicry and severe class imbalance in dermatological images.

5.2 Future Work

Future research should focus on the following priorities:

1. Re-run UgvNet-DS with validation-only checkpoint selection.
2. Use patient-isolated splits to verify leakage prevention.

3. Repeat experiments across at least three controlled seeds to estimate uncertainty.
4. Ablate fusion and attention components in UGVNet and UgvNet-DS.
5. Validate on HAM10000, ISIC, and PAD-UFES-20 under diverse imaging conditions.
6. Benchmark compressed models on edge devices for speed and memory use.

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